

FORM TP 2006017

TEST CODE **01234020**

JANUARY 2006

**CARIBBEAN EXAMINATIONS COUNCIL
SECONDARY EDUCATION CERTIFICATE
EXAMINATION
MATHEMATICS**

Paper 02 – General Proficiency

2 hours 40 minutes

04 JANUARY 2006 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. Answer ALL questions in Section I, and ANY TWO in Section II.
2. Write your answers in the booklet provided.
3. All working must be shown clearly.
4. A list of formulae is provided on page 2 of this booklet.

Examination Materials

Electronic calculator (non-programmable)
Geometry set
Mathematical tables (provided)
Graph paper (provided)

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

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2. (a) Given that $m = -2$ and $n = 4$, calculate the value of $(2m + n)(2m - n)$. (2 marks)
- (b) Solve the simultaneous equations

$$5x + 6y = 37$$

$$2x - 3y = 4.$$

(4 marks)

- (c) Factorise completely

(i) $4x^2 - 25$

(2 marks)

(ii) $6p - 9ps + 4q - 6qs$

(2 marks)

(iii) $3x^2 + 4x - 4.$

(2 marks)

Total 12 marks

3. (a) Given the formula $s = \frac{1}{2}(u + v)t$, express u in terms of v , s , and t . (3 marks)

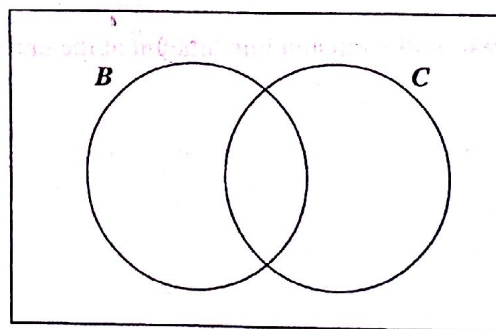
- (b) On a certain day, 300 customers visited a bakery that sells bread and cakes.

70 customers bought cakes only

80 customers bought neither bread nor cakes

$2x$ customers bought bread only

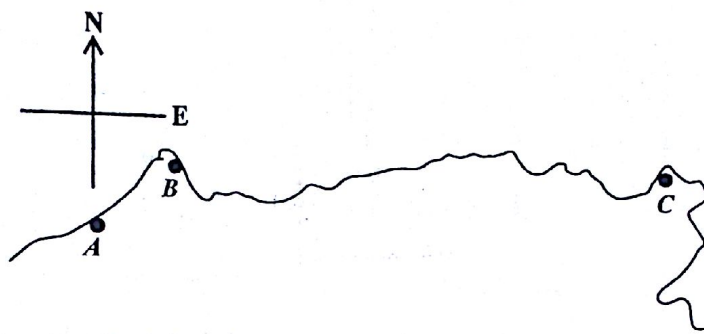
x customers bought both bread and cakes



- (i) U represents the set of customers visiting the bakery on that day, B represents the set of customers who bought bread, and C represents the set of customers who bought cake. Copy and complete the Venn diagram to illustrate the information. (3 marks)
- (ii) Write an expression in x to represent the TOTAL number of customers who visited the bakery on that day. (2 marks)
- (iii) Calculate the number of customers who bought bread ONLY. (3 marks)

Total 11 marks**GO ON TO THE NEXT PAGE**

4. (a) The equation of the line l is $y = 4x + 5$.
- (i) State the gradient of any line that is parallel to l . (1 mark)
- (ii) Determine the equation of the line parallel to l that passes through the point $(2, -6)$. (3 marks)
- (b) An answer sheet is provided for this question.



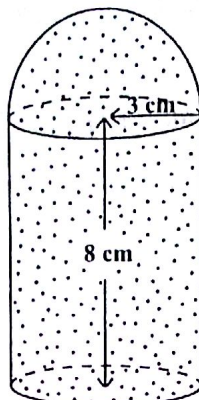
The diagram above shows the positions of three cities, A, B and C, on the north coast of Africa. The scale of the map is 1 : 20 000 000.

Use your answer sheet when answering the following questions. Show all lines and angles used in your calculations.

- (i) Measure and state, in centimetres, the length of the line segment, BC .
(2 marks)
- (ii) Hence, calculate in kilometres, the actual **shortest** distance from City B to City C .
(2 marks)
- (iii) Using a protractor, determine the bearing of B from A .
(4 marks)

Total 12 marks

5. The curved surface area of a cylinder $= 2\pi rh$, where r is the radius and h is the height, and the surface area of a sphere is $4\pi r^2$.



The diagram above, **not drawn to scale**, shows a solid glass paperweight which consists of a hemisphere mounted on a cylinder.

The radius of the hemisphere is 3 cm, the radius of the cylinder is 3 cm and its height is 8 cm.

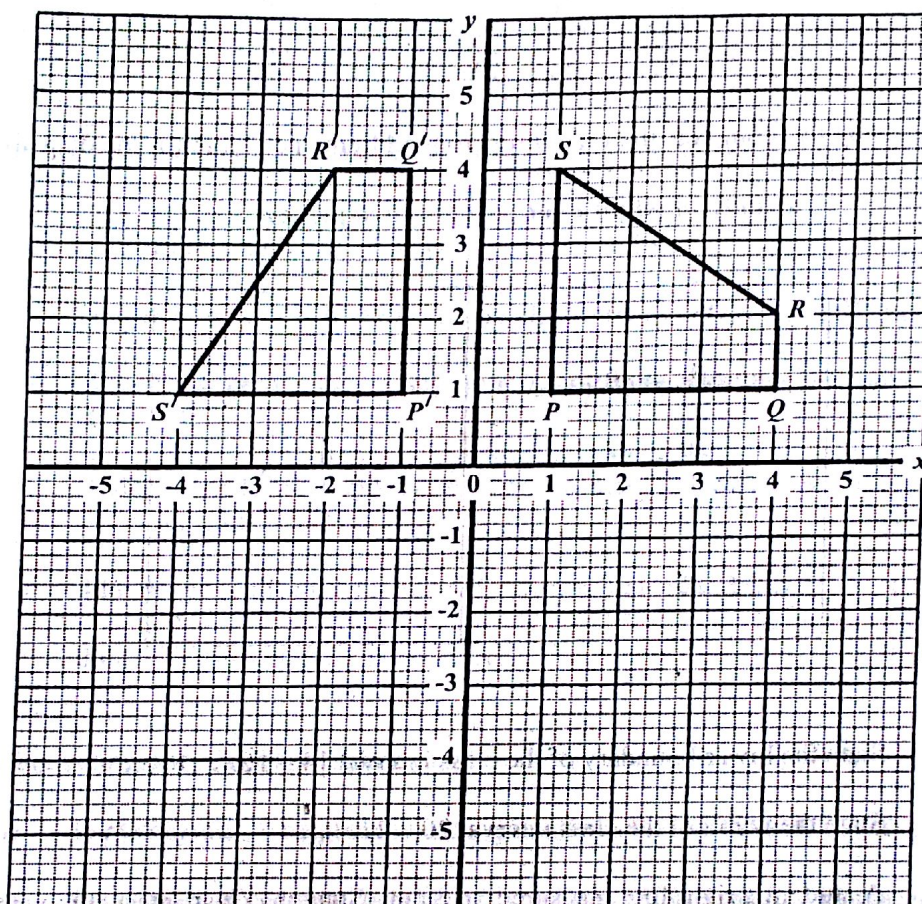
- (a) Calculate, using $\pi = 3.14$,
- the curved surface area of the cylinder (2 marks)
 - the surface area of the hemisphere (2 marks)
 - the TOTAL surface area of the solid paperweight. (2 marks)
- (b) Using a ruler, a pencil, and a pair of compasses, construct the parallelogram $KLMN$, in which $KL = 8$ cm, $KN = 6$ cm, and $\angle LKN = 60^\circ$. (5 marks)

Total 11 marks

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6. An answer sheet is provided for this question.

The diagram below shows quadrilateral $PQRS$, and its image, quadrilateral $P'Q'R'S'$, after it has been rotated.



- State the coordinates of the points R' and S' . (2 marks)
- Describe the rotation completely. (3 marks)
- On the answer sheet provided, draw and label quadrilateral $P''Q''R''S''$ which is the image of $P'Q'R'S'$ after it has been reflected in the x -axis. (3 marks)
- Describe completely, the single transformation that maps quadrilateral $PQRS$ onto $P''Q''R''S''$. (2 marks)

Total 10 marks

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7. The data below are the lengths, to the nearest centimetre, of the right foot of the 25 students in a class.

14	18	20	22	24
15	18	20	22	25
16	18	21	22	25
16	19	22	23	26
17	19	22	23	27

- (a) Copy and complete the following grouped frequency table for the data above.

Length of Right Foot (cm)	Frequency
14 – 16	4
17 – 19	—
20 – 22	8
—	5
26 – 28	2

(2 marks)

- (b) State the lower boundary of the class interval 14 – 16. (1 mark)
- (c) State the width of the class interval 20 – 22. (1 mark)
- (d) A student's right foot measured 16.8 cm. State the class interval in which this length would lie. (1 mark)
- (e) A student was chosen at random from the group, and the length of his right foot was measured. Calculate the probability that the length was GREATER THAN or EQUAL to 20 cm. (2 marks)
- (f) State the modal length of a student's right foot. (1 mark)
- (g) Calculate an estimate of the mean length of a student's right foot using the midpoints of the class intervals in (a) above. (3 marks)

Total 11 marks

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8. The path of a ball thrown in the air is given by the equation $h = 20t - 5t^2$ where h is the vertical distance above the ground (in metres) and t is the time (in seconds) after the ball was thrown.

The table below shows some values of t and the corresponding values of h , correct to 1 decimal place.

t	0	0.5	1	1.5	2	2.5	3	3.5	4
h	0.0	8.8	15	18.8		18.8		8.8	0.0

- (a) Copy and complete the table of values. (2 marks)
- (b) Draw the graph of $h = 20t - 5t^2$ for $0 \leq t \leq 4$. You may proceed as follows:
- (i) Using a scale of 2 cm to represent 0.5 seconds, draw the horizontal t -axis for $0 \leq t \leq 4$. (1 mark)
 - (ii) Using a scale of 1 cm to represent 1 metre, draw the vertical h -axis for $0.0 \leq h \leq 21.0$. (1 mark)
 - (iii) Plot the points from your table of values on the axes drawn. (2 marks)
 - (iv) Join the points with a smooth curve. (1 mark)
- (c) Using your graph, calculate estimates of
- (i) the GREATEST height above the ground reached by the ball (1 mark)
 - (ii) the number of seconds for which the ball was MORE than 12 metres above the ground (2 marks)
 - (iii) the interval of time during which the ball was moving upwards. (1 mark)

Total 11 marks

SECTION II

Answer TWO questions in this section

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) Solve the pair of simultaneous equations

$$2x + y = 7$$

$$x^2 - xy = 6.$$

(6 marks)

- (b) Express $4x^2 - 12x - 3$ in the form $a(x + h)^2 + k$, where a , h and k are real numbers.
(3 marks)

- (c) Using your answer from (b) above, or otherwise, calculate

(i) the minimum value of $4x^2 - 12x - 3$ (1 mark)

(ii) the value of x for which the minimum occurs (1 mark)

(iii) the values of x for which $4x^2 - 12x - 3 = 0$ expressing your answers to 3 significant figures. (4 marks)

Total 15 marks

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10. (a) A shop stocks x Sonix and y Zent radios. It has shelf space for up to 20 radios.

(i) Write an inequality to represent this information. (1 mark)

The owner of the shop spends \$150 to purchase each Sonix radio and \$300 for each Zent radio, she has \$4 500 to spend on the purchase of these radios.

(ii) Write an inequality to represent this information. (1 mark)

The owner of the shop decides to stock at least 6 Sonix and at least 6 Zent radios.

(iii) Write TWO inequalities to represent this information. (2 marks)

- (b) (i) Using a scale of 2 cm to represent 5 Sonix radios and 2 cm to represent 5 Zent radios, draw the horizontal axis for $0 \leq x \leq 30$ and the vertical axis for $0 \leq y \leq 25$. (1 mark)

(ii) On these axes, draw the four boundary lines for the four inequalities written in (a) (i), (ii) and (iii) above. (4 marks)

(iii) Shade the region on your graph that satisfies ALL four of the inequalities written in (a) (i), (ii) and (iii) above. (1 mark)

(iv) State the coordinates of the vertices of the shaded region. (2 marks)

- (c) The owner of the shop sells the radios to make a profit of \$80 on each Sonix and \$100 on each Zent radio.

(i) Express the TOTAL profit in terms of x and y . (1 mark)

(ii) Calculate the MAXIMUM profit. (2 marks)

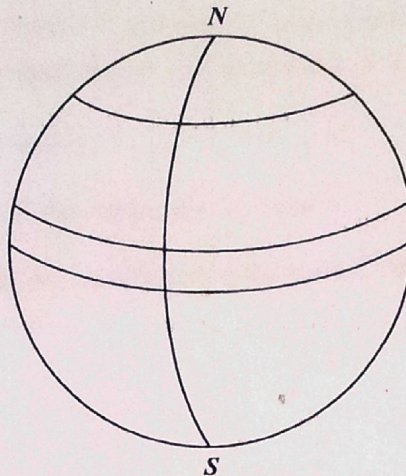
Total 15 marks

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12. In this question assume that the earth is a sphere of radius 6 370 km.

The diagram below shows a sketch of the earth with the north pole, N , and the south pole, S , labelled.



The four arcs on the diagram represent the equator, the Greenwich Meridian, latitude 6°N and latitude 52°N .

- (a) Sketch the diagram and label the

(i) — equator

(ii) Greenwich Meridian

(iii) latitude 52°N

(iv) latitude 6°N .

(6 marks)

- (b) The Greenwich Meridian passes through London (52°N , 0°) and Accra (6°N , 0°).

(i) Show on your diagram the position, L, of London and A, of Accra.

(2 marks)

(ii) Calculate, to the NEAREST kilometre, the **shortest** distance between London and Accra along their common circle of longitude. Use $\pi = 3.14$.

(4 marks)

- (c) Tropical Storm Kyle was reported to be located 5 470 km due west of Accra.

(i) Show on your diagram a possible point of location of Kyle, K. (1 mark)

(ii) Calculate the radius of the circle of latitude on which K lies. (2 marks)

Total 15 marks

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14. (a) (i) Find the inverse of the matrix $M = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$. (2 marks)

(ii) Calculate the values of x and y for which $M \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 1 \end{bmatrix}$. (4 marks)

(b) Under a transformation T , represented by the matrix $\begin{bmatrix} p & q \\ r & s \end{bmatrix}$, the points $A(-4, 2)$ and $B(-2, 5)$ are mapped onto $A'(-2, 4)$ and $B'(-5, 2)$ respectively.

Using a matrix method,

(i) Determine the values of p , q , r and s . (6 marks)

(ii) Calculate the coordinates of the point C' which is the image of $C(-2, 2)$ under T . (3 marks)

Total 15 marks

END OF TEST